Parag Sarvoday Sahu

Research Internship in 3D Computer Vision & Neural Rendering

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- Master's student at École Polytechnique working on 3D vision, inverse rendering and Gaussian splatting.
- First-author manuscript in preparation, Pacific Graphics 2025 paper, 2nd place SIGGRAPH Asia 2025 3DGS Challenge.
- Core skills: differentiable rendering and PyTorch research code; available 22 March to 31 August 2027.

Education

École Polytechnique (IP Paris), Palaiseau, France — Sep 2026 – Present
MSc&T in Visual and Creative Artificial Intelligence — French Government Scholar (Charpak)
Courses: Image Analysis & Computer Vision, Digital Representation & Analysis of Shapes, Computer Animation, Foundations of ML

Indian Institute of Technology (IIT) Gandhinagar, India — 2022 – 2026
B.Tech in Electrical Engineering, Minor in Computer Science — CPI **9.04**/10, Dean's List
Courses: Computer Vision, Machine Learning, Matrix Methods for ML, Probability & Statistics

Publications

TensoIS: Feed-Forward Tensorial Inverse Subsurface Scattering for Perlin Distributed Heterogeneous Media
A. Tiwari, S. Bhardwaj, Y. Bachwana, **P. S. Sahu**, S. Raman — *Pacific Graphics 2025 (Computer Graphics Forum)*

Manuscript on Multi-View Photometric Stereo with 2D Gaussian Splatting
P. S. Sahu, A. Tiwari, V. Shah, S. Raman — *First author, in preparation for CVPR 2027*

Research Experience

IndiaAI Research Fellow (National Merit Fellowship) — Jan 2026 – Present
IIT Gandhinagar | Inverse Rendering with Gaussian Primitives — Advisor: Prof. Shanmuganathan Raman

- Formulated a novel 2D Gaussian Splatting approach to uncalibrated Multi-View Photometric Stereo (MVPS), jointly recovering geometry and materials without known lighting directions or intensities.
- Recovered surface normals at **6.09°** mean angular error on DiLiGenT-MV and supported view-unaligned captures across varying illuminations.
- Cut the required light captures **10**× (10 instead of 96 per view); each object optimizes in under 1 hour on one GPU, versus about 12 hours for PS-NeRF.

Research Intern, Jodhpur, India — Jul 2025 – Dec 2025
3DVisLab, IIT Jodhpur — Advisor: Prof. Avinash Sharma

- Explored learning-based 3D reflective-symmetry detection as a basis for symmetry-aware mesh compression.
- Applied mesh processing and deep learning to high-fidelity 3D human head reconstruction.

Competitions & Projects

SIGGRAPH Asia 2025 3DGS Challenge — Oct 2025 – Nov 2025
Real-time 3D Gaussian Splatting for novel view synthesis — paragsarvoday/DeadlineDinosaur

- Ranked **2nd of about 50 teams worldwide** with **28.04 dB** mean PSNR over 30 test scenes, using a coarse-to-fine progressive-resolution schedule that trains each scene in under 60 seconds.
- Won a **$2,000 USD** prize and presented our method in person at the SIGGRAPH Asia 2025 3DGS Workshop in Hong Kong, covering our backbone choice and hyperparameter study.

Vision Transformer Inference Engine from Scratch — Mar 2026 – Apr 2026
Pure C++ / Eigen, deployed to the browser with WebAssembly — paragsarvoday/ViT-Inference

- Implemented ViT inference without a deep-learning framework, shipping an on-device web demo.

Languages

English: fluent (working language) · Hindi: native · French: A1 (CEFR), currently studying

IT Skills

Programming: Python, C/C++, WebAssembly, MATLAB
Libraries & Tools: PyTorch, gsplat, Mitsuba 3, Eigen, OpenCV, Git/GitHub, LaTeX

Honors and Awards

- **Charpak Master's Scholarship:** Selective French government award for graduate studies (1 of 19 nationwide). 2026
- **Bipin & Rekha Shah & Prof. DV Pai Scholarships:** Awarded for academic excellence at IIT. 2023, 2024
- **JEE Advanced (Top 1%):** Ranked in the top 1 percentile of over 1 million candidates nationwide. 2022

Interests

Badminton · Reading (public bookshelf ↗)